

diverse structures without significant saturation even at a large scale, and that the majority of those structures are novel with respect to Alex-MP-ICSD.

Moreover, we benchmark MatterGen against previous generative models for materials and show a significant improvement in performance. We focus on two key metrics: (1) the percentage of S.U.N. materials among generated samples, measuring the overall success rate of generating promising candidates, and (2) the average RMSD between generated samples and their DFT-relaxed structures, measuring the distance to equilibrium. We also compare to MatterGen-MP, which is a MatterGen model trained only on MP-20, i.e., the same, smaller, dataset used by the other baselines. In Fig. 2(e-f), MatterGen-MP shows a 1.8 times increase in the percentage of S.U.N. structures and a 3.1 times decrease in average RMSD compared with the previous state-of-the-art CDVAE [4]. When comparing MatterGen with MatterGen-MP, we observe a further 1.6 times increase in the percentage of S.U.N. structures and a 5.5 times decrease in RMSD as a result of scaling up the training dataset.

In summary, we have shown that MatterGen is able to generate S.U.N. materials at a substantially higher rate compared to previous generative models while the generated structures are orders of magnitudes closer to their local energy minimum. Next, we fine-tune the pre-trained base model of MatterGen towards different downstream applications, including target chemistry (Section 2.3), symmetry (Section 2.4), and scalar property constraints (Sections 2.5 and 2.6).

2.3 Generating materials with target chemistry

Finding the most stable material structures in a target chemical system (e.g., Li-Co-O) is crucial to define the true convex hull required for assessing stability, and indeed is one of the major challenges in materials design [48]. The most comprehensive approach for this task is *ab initio* RSS [49], which has been used to discover many novel materials that were later experimentally synthesized [48]. The biggest drawback of RSS is its computational cost, as the thorough exploration of even a ternary compound can require hundreds of thousands of DFT relaxations. In recent years, the combination of generating structures via RSS, substitution or evolutionary methods with MLFFs has proven successful in exploring chemical systems [23, 27, 50, 51]. Here, we evaluate the ability of MatterGen to explore target chemical systems by comparing it with substitution [23] and RSS [49, 52]. We equip all methods with the MatterSim[53] MLFF, which is used to pre-relax and filter the generated structures by their predicted stability before running more expensive DFT calculations. We fine-tune the MatterGen base model (Appendix B.1) and steer the generated structures towards different target chemical systems and an energy above hull of 0.0 eV/atom. We perform the benchmark evaluation for nine ternary, nine quaternary, and nine quinary chemical systems. For each of these three groups, we pick three chemical systems at random from the following categories: well explored, partially explored, and not explored. See Appendix D.4 for additional details. In Fig. 3(a-b) we see that MatterGen generates the highest percentage of S.U.N. structures for every system type and every chemical complexity. As highlighted in Fig. 3(c), MatterGen also finds the highest number of unique structures on the combined convex hull in (1) ‘partially explored’ systems, where existing known

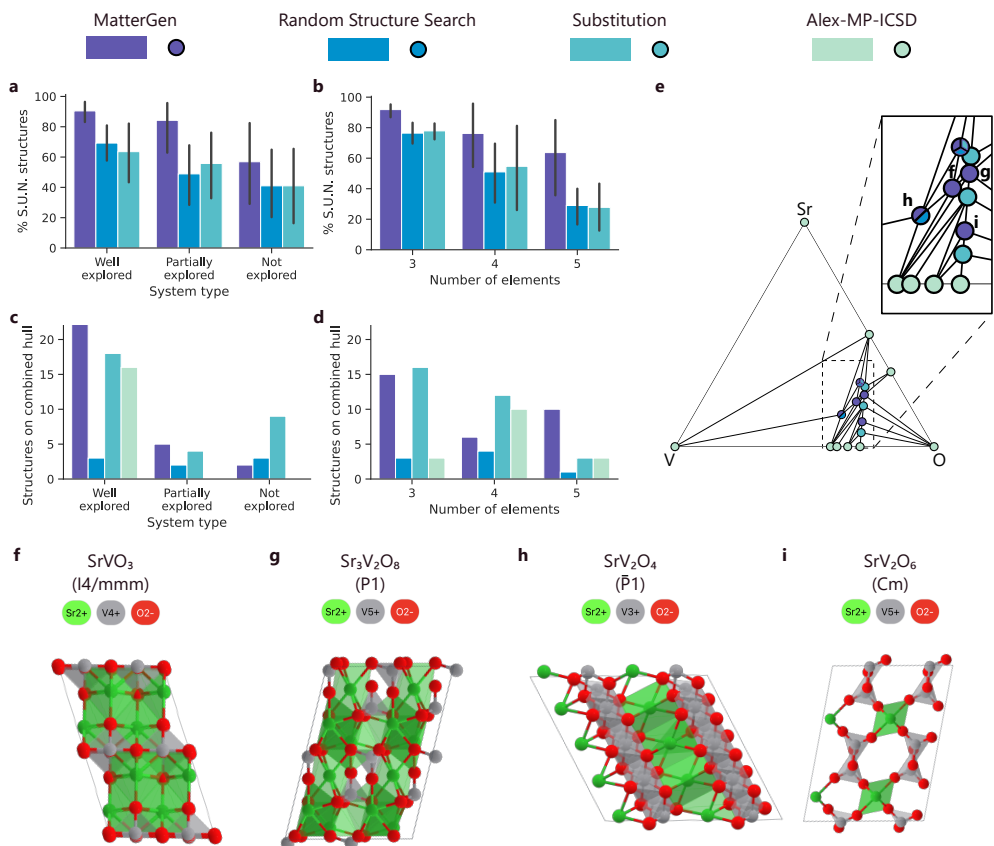


Fig. 3: Generating materials in target chemical system. (a-b) Mean percentage of S.U.N. structures generated by MatterGen and baselines for 27 chemical systems, grouped by system type (a) and number of elements (b). Vertical black lines indicate maximum and minimum values. (c-d) Number of structures on the combined convex hull found by each method and in the Alex-MP-ICSD dataset, grouped by system type (c) and number of elements (d). (e) Convex hull diagram for V-Sr-O, a well-explored ternary system. The dots represent structures on the hull, their coordinates represent the element ratio of their composition, and their color indicates by which method they were discovered. (f-i) Four of the five structures MatterGen discovered on the V-Sr-O hull depicted in (e), along with their composition and space group.

structures near the hull were provided during training, and in (2) ‘well-explored systems’, where structures near the hull are known but were not provided in training. While substitution offers a comparable or more efficient way to generate structures on the hull for ternary and quaternary systems, MatterGen achieves better performance on quinary systems, as shown in Fig. 3(d). Remarkably, the strong performance of MatterGen in quinary systems was achieved with only 10,240 generated samples,

compared to $\sim 70,000$ samples for substitution and 600,000 for RSS. This underscores the enormous efficiency gains that can be realized with generative models by proposing better initial candidates. Finally, in Fig. 3(e) we show that MatterGen finds five novel structures on the combined hull for V-Sr-O—an example of a well-explored ternary system—while substitution finds four, and RSS only two. A few of the structures discovered by MatterGen are shown in Fig. 3(f-i), and are analyzed in-depth in Appendix D.4.2.

2.4 Designing materials with target symmetry

The symmetry of a material directly affects its electronic and vibrational properties, and is a determining factor for its topological [54] and ferroelectric [55] characteristics. The generation of S.U.N. materials with a given symmetry is a challenging task, as the symmetric arrangement of atoms in space is hard to enforce without resorting to explicit constraints based on already known materials. Here, we assess MatterGen’s ability to generate S.U.N. materials with a target symmetry by fine-tuning it on space group labels. We choose 14 space groups at random from the subset of space groups that had at least 1000 entries in the training dataset, two for each of the seven crystal systems, and generate 256 structures per space group. The results are shown in Fig. 4(a). On average, the fraction of generated S.U.N. structures that belong to the target space group is 20%, and still surpassing 10% for some of the most highly symmetric space groups that were chosen, e.g., $P6_3/mmc$ and $Im\bar{3}$. This is a notable result given that most previous generative models struggled in generating highly symmetric crystals [4, 56]. In Fig. 4(b), we show four randomly generated S.U.N. structures from different space groups. Additional details and results are provided in Appendix D.5.

2.5 Designing materials with target magnetic, electronic, and mechanical properties

There is an enormous need for new materials with improved properties across a wide range of real-world applications, e.g., for designing carbon capture technologies, solar cells, or semiconductors [24–26]. The classical screening-based approach starts from a set of candidates and selects the ones with the best properties. However, screening methods are unable to explore structures beyond the set of known materials. Here, we demonstrate MatterGen’s ability to directly generate S.U.N. materials with target constraints on three different single-property inverse design tasks. These feature a diverse set of properties—magnetic, electronic, and mechanical—with varying degrees of available labeled data for fine-tuning the model. In the first task, we aim to generate materials with high magnetic density, a prerequisite for permanent magnets. We fine-tune the model on 605,000 structures with DFT magnetic density labels (calculated assuming ferromagnetic ordering) and then generate structures with a target magnetic density value of $0.20 \text{ \AA}^{-3}$. Second, we search for materials with a specific electronic property. We fine-tune the model on 42,000 structures with DFT band gap labels, then sample materials with a target calculated band gap value of 3.0 eV. Finally, we target structures with high bulk modulus—an important property for superhard materials. We fine-tune the model on only 5,000 labeled structures, and sample with a target

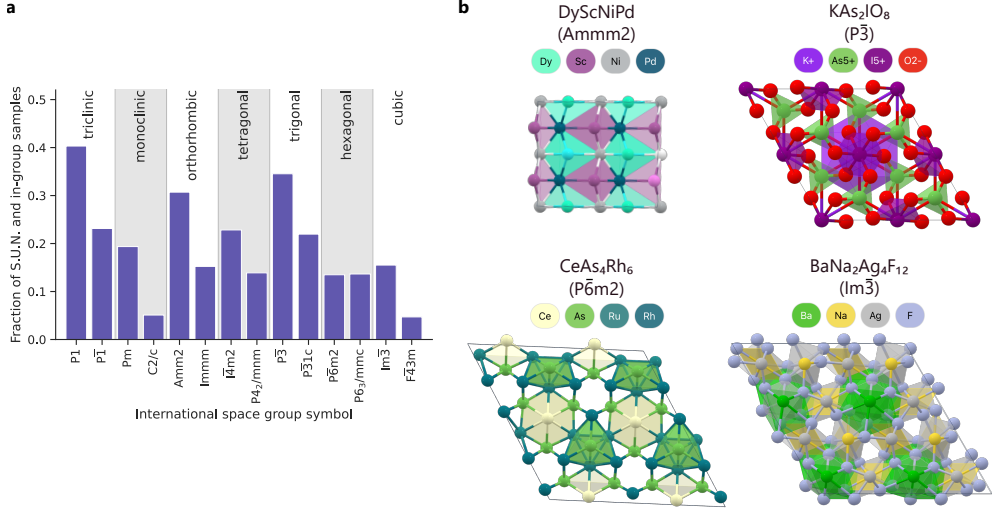


Fig. 4: Generating materials with target symmetry. (a) Fraction of generated S.U.N. structures that belong to the target space group for 14 randomly chosen space groups spanning the seven lattice types. (b) Four randomly selected S.U.N. structures generated by MatterGen, along with their chemical formula and space group.

value of 400 GPa. While the tasks above were chosen to evaluate the generality of the model, we note that additional follow-up investigations would be required to assess the suitability of these materials for specific applications, e.g., a superhard material needs to satisfy additional constraints such as a high shear modulus, or a permanent magnet needs a suitable magnetic order and critical temperature. See Appendix D.6 for more details.

In Fig. 5(a-c), we highlight the significant shift in the distribution of property values among S.U.N. samples generated by MatterGen towards the desired targets, even when the targets are at the tail of the data distribution. In particular, this still holds true for properties where the number of DFT labels available for fine-tuning the model is substantially smaller than the size of the unlabeled training data. In Fig. 5(d-f) we showcase the S.U.N. structures with the best property values generated by MatterGen for each task. See Appendix D.6.2 for additional analysis.

Moreover, we assess how many S.U.N. structures satisfying extreme property constraints can be found by MatterGen when given a limited budget for DFT property calculations. As a baseline, we count the number of materials in the labeled fine-tuning dataset that satisfy the constraint. We also compare with a screening approach, which scans previously unlabeled materials for promising candidates. In contrast to the previous experiment, we fine-tune MatterGen with labels predicted by a machine learning property predictor—the same used for the screening baseline—when the dataset is not fully labeled. As shown in Fig. 5(g), MatterGen is able to find up to 47 S.U.N. structures with magnetic density above 0.2 Å^{-3} , much more than the 26 materials with such high property values in the fine-tuning dataset. Since the dataset is fully

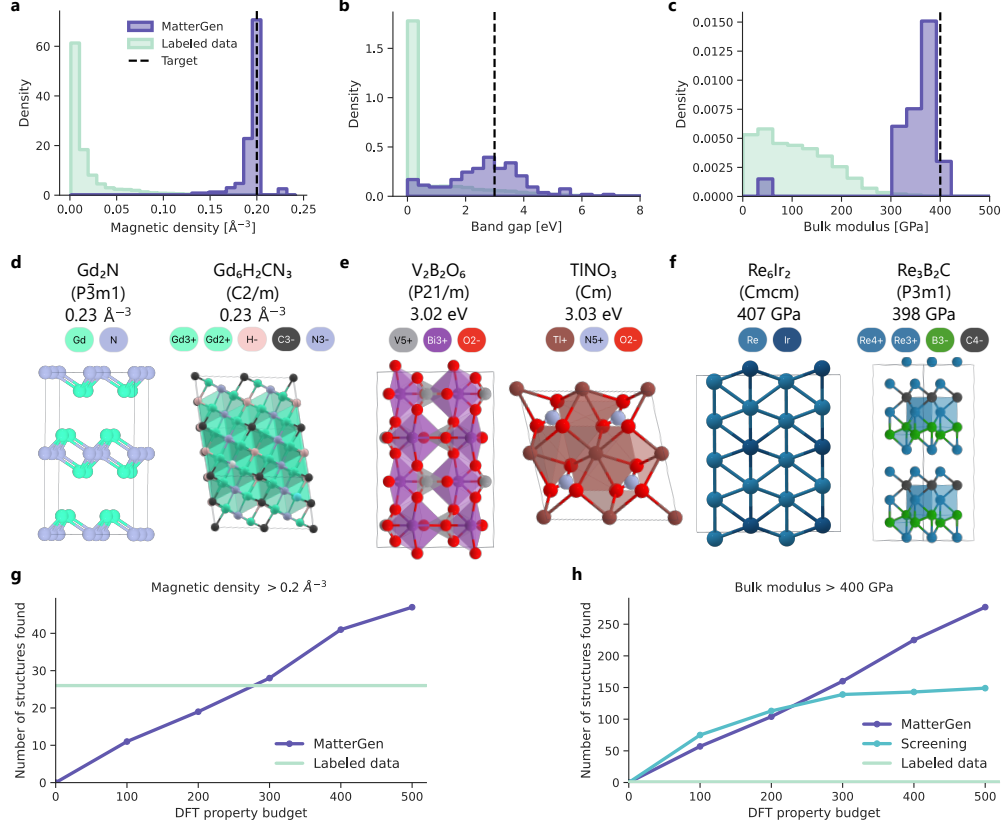


Fig. 5: Generating materials with target magnetic, electronic, and mechanical properties. (a-c) Density of property values among (1) generated S.U.N. samples by MatterGen, and (2) structures in the labeled fine-tuning dataset for a magnetic, electronic, and mechanical property, respectively. The property target for MatterGen is shown as a black dashed line. Magnetic density values $< 10^{-3} \text{ \AA}^{-3}$ in (a) are excluded from the labeled data to improve readability. (d-f) Visualization of S.U.N. structures with the best property values generated by MatterGen for magnetic density (d), band gap (e), and bulk modulus (f). Alongside each structure, the chemical formula, space group and property value is shown. (g-h) Number of S.U.N. structures that satisfy target constraints found MatterGen compared to number of structures found by base-lines across a range of DFT property calculation budgets.

labeled, there is no screening baseline available. In Fig. 5(h), we see that MatterGen finds substantially more S.U.N. materials with high bulk modulus than screening. While the number of structures found by screening saturates with increasing budget, MatterGen keeps discovering S.U.N. structures at an almost constant rate. Given a budget of 500 DFT property calculations, we find 277 S.U.N. structures (with 126 distinct compositions), almost double the number found with a screening approach (149,